

TMA4115 V25

Opg 3, 4, 5, 6, 7, 8, 9, 10

Opg 3

$$A = \begin{bmatrix} 1 & 0 & 4 \\ -2 & 1 & -5 \\ 2 & -2 & z^2-2 \end{bmatrix}$$

A er inverterbar $\Leftrightarrow \det(A) \neq 0$

$$a) \det(A) = 1 \begin{vmatrix} 1 & -5 \\ -2 & z^2-2 \end{vmatrix} + 4 \begin{vmatrix} -2 & 1 \\ 2 & -2 \end{vmatrix}$$

$$\det(A) = z^2 - 2 - (-5 \cdot -2) + 4(4 - 2)$$

$$\det(A) = z^2 - 2 - 10 + 4 \cdot 2$$

$$\det(A) = z^2 - 2 - 10 + 8$$

$$\det(A) = z^2 - 4 = (z-2)(z+2)$$

Vi ser at $\det(A) = 0$ for ± 2 .

A er inverterbar for alle $z \neq \pm 2$

$$b) Ax = \begin{bmatrix} 1 \\ 0 \\ z-4 \end{bmatrix}$$

$$\left[\begin{array}{ccc|c} 1 & 0 & 4 & 1 \\ -2 & 1 & -5 & 0 \\ 2 & -2 & z^2-2 & z-4 \end{array} \right] \xrightarrow{R_2 + 2R_1, R_3 - 2R_1} \left[\begin{array}{ccc|c} 1 & 0 & 4 & 1 \\ 0 & 1 & 3 & 2 \\ 0 & -2 & z^2-10 & z-6 \end{array} \right]$$

$$\xrightarrow{R_3 + 2R_2} \left[\begin{array}{ccc|c} 1 & 0 & 4 & 1 \\ 0 & 1 & 3 & 2 \\ 0 & 0 & z^2-4 & z-2 \end{array} \right]$$

For $z=2 \Rightarrow \begin{matrix} z^2-4 = 2^2-4 = 0 \\ 4-4 = 0 \\ 0=0 \end{matrix} \Rightarrow \text{Nullrad} \Rightarrow \text{uendelig mange løsninger}$

For $z=-2 \Rightarrow \begin{matrix} (-2)^2-4 = 0 \\ 4-4 = 0 \\ 0=0 \end{matrix} \Rightarrow \text{Ingen løsning}$

For $z \neq \pm 2 \Rightarrow 1 \text{ løsning}$

Opg 4

$$U = \{A \in M_2(\mathbb{R}) \mid A^T = -A\}$$

a) Vi må sjekke at nullmatrise er med i U , og at U er lukket under addisjon og skælmultiplikasjon.

$$i) \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}^T = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = -\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \quad \checkmark$$

$$ii) \text{ L2 } A, B \in U. \text{ D2 er } A^T = -A \text{ og } B^T = -B$$

$$(A+B)^T = A^T + B^T = -A + (-B) = -(A+B) \quad \checkmark$$

$$iii) \text{ L2 } A \in U. \text{ D2 h2r vi for } k \in \mathbb{R} \text{ at } (kA)^T = kA^T = k(-A) = -(kA) \quad \checkmark$$

$$b) \text{ L2 } A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \in U$$

$$A^T = \begin{bmatrix} a & c \\ b & d \end{bmatrix} = \begin{bmatrix} -a & -b \\ -c & -d \end{bmatrix} = -A$$

$$a = -a, c = -b, d = -d$$

$$\text{For at } a = -a \text{ og } d = -d, \text{ m2 } a = d = 0$$

$$\begin{bmatrix} 0 & b \\ -b & 0 \end{bmatrix} = b \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$$

$$B = \left\{ \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \right\}$$

Siden basisen bare h2r en matrise, er dimensionen

$$\dim(U) = 1$$

Underrom

- Nullmatrise
- Addisjon
- Skælmultiplikasjon

$$A^T = \begin{bmatrix} a & c \\ b & d \end{bmatrix}$$

Opg 5

$$T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$$

$$T\left(\begin{pmatrix} 0 \\ 2 \\ 1 \end{pmatrix}\right) = \begin{pmatrix} 2 \\ 1 \\ 3 \end{pmatrix} \quad T\left(\begin{pmatrix} 0 \\ 2 \\ 2 \end{pmatrix}\right) = \begin{pmatrix} 4 \\ 4 \\ 2 \end{pmatrix} \quad T\left(\begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix}\right) = \begin{pmatrix} 2 \\ 0 \\ 2 \end{pmatrix}$$

Vi ønsker å finne $T(x) = Ax$ og vi har fått oppgitt at

$$AX = A \begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 1 \\ 1 & 2 & 1 \end{pmatrix} = \begin{pmatrix} 2 & 4 & 2 \\ 1 & 4 & 0 \\ 3 & 2 & 2 \end{pmatrix} \leftarrow \text{Kaller for } Y$$

$$\begin{aligned} AX &= Y \\ AX X^{-1} &= Y X^{-1} \\ AI &= Y X^{-1} \\ A &= Y X^{-1} \end{aligned}$$

Standardmetrise

$$T(x) = Ax$$

$$A = YX^{-1}$$

$$X^{-1} = \left[\begin{array}{ccc|ccc} 0 & 0 & 2 & 1 & 0 & 0 \\ 2 & 2 & 1 & 0 & 1 & 0 \\ 1 & 2 & 1 & 0 & 0 & 1 \end{array} \right] \xrightarrow{R_2 \leftrightarrow R_1} \left[\begin{array}{ccc|ccc} 1 & 2 & 1 & 0 & 0 & 1 \\ 2 & 2 & 1 & 0 & 1 & 0 \\ 0 & 0 & 2 & 1 & 0 & 0 \end{array} \right] \xrightarrow{R_2 - 2R_1} \left[\begin{array}{ccc|ccc} 1 & 2 & 1 & 0 & 0 & 1 \\ 0 & -2 & -1 & 0 & 1 & -2 \\ 0 & 0 & 2 & 1 & 0 & 0 \end{array} \right]$$

$$\xrightarrow{-\frac{1}{2}R_2} \left[\begin{array}{ccc|ccc} 1 & 2 & 1 & 0 & 0 & 1 \\ 0 & 1 & 1/2 & 0 & -1/2 & 1 \\ 0 & 0 & 2 & 1 & 0 & 0 \end{array} \right] \xrightarrow{\frac{1}{2}R_3} \left[\begin{array}{ccc|ccc} 1 & 2 & 1 & 0 & 0 & 1 \\ 0 & 1 & 1/2 & 0 & -1/2 & 1 \\ 0 & 0 & 1 & 1/2 & 0 & 0 \end{array} \right]$$

$$\xrightarrow{R_2 - \frac{1}{2}R_3} \left[\begin{array}{ccc|ccc} 1 & 2 & 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & -1/4 & -1/2 & 1 \\ 0 & 0 & 1 & 1/2 & 0 & 0 \end{array} \right] \xrightarrow{R_1 - R_3} \left[\begin{array}{ccc|ccc} 1 & 2 & 0 & -1/2 & 0 & 1 \\ 0 & 1 & 0 & -1/4 & -1/2 & 1 \\ 0 & 0 & 1 & 1/2 & 0 & 0 \end{array} \right]$$

$$\xrightarrow{R_1 - 2R_2} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 0 & 1 & -1 \\ 0 & 1 & 0 & -1/4 & -1/2 & 1 \\ 0 & 0 & 1 & 1/2 & 0 & 0 \end{array} \right] \quad \begin{array}{ccc} 0 & 1 & -1 \\ -1/4 & -1/2 & 1 \\ 1/2 & 0 & 0 \end{array}$$

$$A = YX^{-1} = \begin{pmatrix} 2 & 4 & 2 \\ 1 & 4 & 0 \\ 3 & 2 & 2 \end{pmatrix} \begin{pmatrix} 0 & 1 & -1 \\ -1/4 & -1/2 & 1 \\ 1/2 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 2 & 4 & 2 \\ 1 & 4 & 0 \\ 3 & 2 & 2 \end{pmatrix} \begin{pmatrix} 0 & 0 & 2 \\ -1 & -1 & 3 \\ 1/2 & 2 & -1 \end{pmatrix}$$

$$\Rightarrow \text{Standardmetrisen} = \begin{pmatrix} 0 & 0 & 2 \\ -1 & -1 & 3 \\ 1/2 & 2 & -1 \end{pmatrix}$$

$$\det(A) = 2 \cdot \begin{vmatrix} -1 & -1 \\ 1/2 & 2 \end{vmatrix} = 2(-2 + 1/2) = -3 \neq 0$$

Siden $\det(A) \neq 0$ er A invertierbar og lineærtransformasjonen er både injektiv og surjektiv

Opg 6

b) $\langle x, y \rangle = x_1 y_1 + 4x_2 y_2 + 9x_3 y_3$

$$v_1 = \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix} \quad v_2 = \begin{bmatrix} -1 \\ 2 \\ -1 \end{bmatrix}$$

Gram-Schmidt

$$u_1 = v_1$$

$$u_2 = v_2 - \text{proj}_{u_1}(v_2)$$

$$\text{der } \text{proj}_{u_1}(v_2) = \frac{\langle v_2, u_1 \rangle}{\langle u_1, u_1 \rangle} u_1$$

$$u_1 = v_1 = \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix}$$

$$\langle v_2, u_1 \rangle = -1 \cdot 2 + 4 \cdot 2 \cdot (-1) + 9 \cdot (-1) \cdot 1 = -2 - 8 - 9 = -19$$

$$\langle u_1, u_1 \rangle = 2^2 + 4(-1)^2 + 9(1)^2 = 4 + 4 + 9 = 17$$

$$u_2 = v_2 - \frac{\langle v_2, u_1 \rangle}{\langle u_1, u_1 \rangle} u_1$$

$$u_2 = \begin{bmatrix} -1 \\ 2 \\ -1 \end{bmatrix} - \frac{-19}{17} \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix}$$

$$u_2 = \begin{bmatrix} -1 \\ 2 \\ -1 \end{bmatrix} + \frac{19}{17} \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix}$$

$$u_2 = \begin{bmatrix} -1 + \frac{38}{17} \\ 2 - \frac{19}{17} \\ -1 + \frac{19}{17} \end{bmatrix} = \begin{bmatrix} 21/17 \\ 15/17 \\ 2/17 \end{bmatrix} = \begin{bmatrix} 21 \\ 15 \\ 2 \end{bmatrix}$$

En ortogonal basis er $\left\{ \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix}, \begin{bmatrix} 21 \\ 15 \\ 2 \end{bmatrix} \right\}$

Oppgave 7

a) $A = \begin{bmatrix} a & b-a \\ 0 & b \end{bmatrix}$ hvor $a, b \in \mathbb{R}$ og $a \neq b$

Dette er en øvre triangulær matrise, fordi alt under diagonalen er 0

$$\lambda_1 = a \quad \lambda_2 = b$$

Finn egenvektor til $\lambda_1 = a$

$$(A - aI)x = 0$$

$$\left(\begin{bmatrix} a & b-a \\ 0 & b \end{bmatrix} - \begin{bmatrix} a & 0 \\ 0 & a \end{bmatrix} \right) x = 0$$

$$\begin{bmatrix} 0 & b-a \\ 0 & b-a \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0$$

$$0x_1 + (b-a)x_2 = 0$$

$$(b-a)x_2 = 0$$

$$b \neq a \Rightarrow x_2 = 0$$

$$x_1 \text{ er fri} \Rightarrow \text{velger } x_1 = 1$$

$$v_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

Finn egenvektor til $\lambda_2 = b$

$$(A - bI)x = 0$$

$$\left(\begin{bmatrix} a & b-a \\ 0 & b \end{bmatrix} - \begin{bmatrix} b & 0 \\ 0 & b \end{bmatrix} \right) x = 0$$

$$\begin{bmatrix} a-b & b-a \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0$$

$$(a-b)x_1 + (b-a)x_2 = 0$$

$$(a-b)x_1 - (a-b)x_2 = 0$$

$$(a-b)(x_1 - x_2) = 0$$

$$\text{Siden } a \neq b \Rightarrow a-b \neq 0 \Rightarrow x_1 - x_2 = 0 \Rightarrow x_1 = x_2 \quad \text{Velg } x_2 = 1 \Rightarrow x_1 = 1$$

$$v_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

For triangulære matriser er egenverdierne tallene på diagonalen

L2g P og D

$$\lambda_1 = 2, v_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$\lambda_2 = b, v_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$P = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \quad D = \begin{pmatrix} 2 & 0 \\ 0 & b \end{pmatrix}$$

$$A = PDP^{-1}$$

Finn P^{-1}

$$P = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$$

$$\left[\begin{array}{cc|cc} 1 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{array} \right] \xrightarrow{R_1 - R_2} \left[\begin{array}{cc|cc} 1 & 0 & 1 & -1 \\ 0 & 1 & 0 & 1 \end{array} \right] \Rightarrow P^{-1} = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix}$$

Finn A^k

$$A = PDP^{-1}$$

$$A^k = PD^kP^{-1}$$

$$D^k = \begin{pmatrix} 2^k & 0 \\ 0 & b^k \end{pmatrix}$$

$$A^k = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 2^k & 0 \\ 0 & b^k \end{pmatrix} \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix}$$

$$A^k = \begin{pmatrix} 2^k & b^k \\ 0 & b^k \end{pmatrix} \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix}$$

$$A^k = \begin{pmatrix} 2^k & b^k - 2^k \\ 0 & b^k \end{pmatrix}$$

$$\begin{array}{c|cc} & 2^k & 0 \\ \hline & 0 & b^k \\ \hline 1 & 1 & 2^k & b^k \\ 0 & 1 & 0 & b^k \end{array}$$

$$\begin{array}{c|cc} & 1 & -1 \\ \hline & 0 & 1 \\ \hline 2^k & b^k & 2^k & b^k - 2^k \\ 0 & b^k & 0 & b^k \end{array}$$

b) $y' = \begin{bmatrix} -2 & 5 \\ 0 & 3 \end{bmatrix} y \quad y_1(0) = 2 \quad y_2(0) = -4$

Finn egenverdier

$$A = \begin{pmatrix} -2 & 5 \\ 0 & 3 \end{pmatrix} \text{ er øvre triangulær}$$

$$\lambda_1 = -2, \lambda_2$$

D er en diagonalmatrix
med egenverdier på diagonalen

Kolonnene i P er
egenvektorene

Finn egenvektorer

Egenvektor til $\lambda_1 = -2$

$$(A - \lambda I)x = 0 \Rightarrow (A + 2I)x = 0$$

$$\left(\begin{bmatrix} -2 & 5 \\ 0 & 3 \end{bmatrix} + \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} \right) x = 0$$

$$\begin{bmatrix} 0 & 5 \\ 0 & 5 \end{bmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \Rightarrow \begin{matrix} x_2 = 0 \\ x_1 = \text{fri} \Rightarrow \text{sett } 1 \end{matrix}$$

$$v_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

Egenvektor til $\lambda_2 = 3$

$$(A - 3I)x = 0$$

$$\left(\begin{bmatrix} -2 & 5 \\ 0 & 3 \end{bmatrix} - \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix} \right) x = 0$$

$$\begin{bmatrix} -5 & 5 \\ 0 & 0 \end{bmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$-5x_1 + 5x_2 = 0 \Rightarrow -x_1 + x_2 = 0 \Rightarrow x_1 = x_2 \quad \text{Velg } x_1 = 1 \Rightarrow x_2 = 1$$

$$v_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

Lsg generell løsning

$$y(t) = c_1 v_1 e^{\lambda_1 t} + c_2 v_2 e^{\lambda_2 t}$$

$$y(t) = c_1 \begin{pmatrix} 1 \\ 0 \end{pmatrix} e^{-2t} + c_2 \begin{pmatrix} 1 \\ 1 \end{pmatrix} e^{3t}$$

Bruk initislbetingelser

$$y_1(0) = 2, y_2(0) = -4$$

$$y(0) = c_1 \begin{pmatrix} 1 \\ 0 \end{pmatrix} e^{-2 \cdot 0} + c_2 \begin{pmatrix} 1 \\ 1 \end{pmatrix} e^{3 \cdot 0} = c_1 \begin{pmatrix} 1 \\ 0 \end{pmatrix} + c_2 \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$y(0) = \begin{pmatrix} c_1 + c_2 \\ c_2 \end{pmatrix} = \begin{pmatrix} 2 \\ -4 \end{pmatrix} \Rightarrow \begin{matrix} c_1 + c_2 = 2 \\ c_2 = -4 \end{matrix} \Rightarrow c_1 = 6$$

$$y(t) = 6 \begin{pmatrix} 1 \\ 0 \end{pmatrix} e^{-2t} - 4 \begin{pmatrix} 1 \\ 1 \end{pmatrix} e^{3t}$$

Opg 9

$$S = \{p_1(x), p_2(x), p_3(x)\}$$

$$p_1(x) = x^2 + 1, p_2(x) = 6x^2 + x + 2, p_3(x) = 3x^2 + x$$

Vi skal finne koordinatvektorene til

$$q(x) = x^2 + 2x + 3$$

Hvz betyr dette?

Vi skal skrive $q(x) = a p_1(x) + b p_2(x) + c p_3(x)$
og finne tallene a, b, c

Sett opp uttrykket

$$x^2 + 2x + 3 = a(x^2 + 1) + b(6x^2 + x + 2) + c(3x^2 + x)$$

Gang inn

$$a(x^2 + 1) = ax^2 + a$$

$$b(6x^2 + x + 2) = 6bx^2 + bx + 2b$$

$$c(3x^2 + x) = 3cx^2 + cx$$

Samle ledd

$$x^2: ax^2 + 6bx^2 + 3cx^2 = (a + 6b + 3c)x^2$$

$$x: bx + cx = (b + c)x$$

$$\text{konstantledd: } a + 2b$$

$$\text{Vi får: } x^2 + 2x + 3 = (a + 6b + 3c)x^2 + (b + c)x + (a + 2b)$$

Sammenlign koeffisienter

$$x^2: a + 6b + 3c = 1$$

$$x: b + c = 2$$

$$\text{konstantledd: } a + 2b = 3$$

Løs systemet

$$\left[\begin{array}{ccc|c} 1 & 6 & 3 & 1 \\ 0 & 1 & 1 & 2 \\ 1 & 2 & 0 & 3 \end{array} \right] \xrightarrow{R_3 - R_1} \left[\begin{array}{ccc|c} 1 & 6 & 3 & 1 \\ 0 & 1 & 1 & 2 \\ 0 & -4 & -3 & 2 \end{array} \right] \xrightarrow{R_3 + 4R_2} \left[\begin{array}{ccc|c} 1 & 6 & 3 & 1 \\ 0 & 1 & 1 & 2 \\ 0 & 0 & 1 & 10 \end{array} \right]$$

$$\begin{array}{l} R_1 - 3R_3 \\ R_2 - R_3 \end{array} \left[\begin{array}{ccc|c} 1 & 6 & 0 & -29 \\ 0 & 1 & 0 & -8 \\ 0 & 0 & 1 & 10 \end{array} \right] \xrightarrow{R_1 - 6R_2} \left[\begin{array}{ccc|c} 1 & 0 & 0 & 19 \\ 0 & 1 & 0 & -8 \\ 0 & 0 & 1 & 10 \end{array} \right]$$

$$[q]_S = \begin{bmatrix} 19 \\ -8 \\ 10 \end{bmatrix} \quad \text{Dette er koordinatvektoren til } q(x) \text{ i basisen } S$$

Oppeve 8

$$y'' + 5y' + 6y = e^{-2t}, \quad y(0) = 1, \quad y'(0) = 1$$

Lineær 2.ordens differensligning

Total løsning

$$y = y_h + y_p$$

- y_h = homogen løsning
- y_p = partikulær løsning

Løs homogen ligning

$$y'' + 5y' + 6y = 0$$

Karakteristisk ligning:

$$r^2 + 5r + 6 = 0$$

$$(r+2)(r+3) = 0$$

$$r = -2 \quad \vee \quad r = -3$$

Homogen løsning er:

$$y_h = c_1 e^{-2t} + c_2 e^{-3t}$$

Finn partikulær løsning

Høgresiden er e^{-2t}

Venligvis ville vi gjettest $y_p = Ae^{-2t}$

Men e^{-2t} finnes allerede i homogen løsning

Derfor må vi gange med t

$$y_p = At e^{-2t}$$

Deriver y_p

$$y_p = \underbrace{At}_u \underbrace{e^{-2t}}_v$$

$$y_p' = u' \cdot v + u \cdot v'$$

$$y_p' = A e^{-2t} - \underbrace{2At e^{-2t}}_{u' \cdot v}$$

$$y_p'' = -2A e^{-2t} - u' \cdot v + u \cdot v'$$

$$y_p'' = -2A e^{-2t} - 2A e^{-2t} + 4A t e^{-2t}$$

$$y_p'' = -4A e^{-2t} + 4A t e^{-2t}$$

Sett inn i ligningen

$$y'' + 5y' + 6y = e^{-2t}$$

$$(-4A e^{-2t} + 4A t e^{-2t}) + 5(A e^{-2t} - 2A t e^{-2t}) + 6(At e^{-2t}) = e^{-2t}$$

$$A e^{-2t} = e^{-2t}$$

$$A = 1$$

$$\text{Så } y_p = t e^{-2t}$$

Generell løsning

$$y = y_h + y_p$$

$$y = c_1 e^{-2t} + c_2 e^{-3t} + t e^{-2t}$$

Bruk $y(0) = 1$

$$y(0) = c_1 e^0 + c_2 e^0 + 0 \cdot e^0$$

$$y(0) = c_1 + c_2$$

$$c_1 + c_2 = 1$$

$$c_1 = 1 - c_2$$

Brug $y'(0) = 1$

$$y = c_1 e^{-2t} + c_2 e^{3t} + t e^{-2t}$$

$$y' = -2c_1 e^{-2t} - 3c_2 e^{3t} + (e^{-2t} - 2t e^{-2t})$$

$$y'(0) = -2c_1 - 3c_2 + 1$$

$$-2c_1 - 3c_2 + 1 = 1$$

$$-2c_1 - 3c_2 = 0$$

$$\text{Vi h dte } c_1 = 1 - c_2$$

$$-2(1 - c_2) - 3c_2 = 0$$

$$-2 + 2c_2 - 3c_2 = 0$$

$$-2 - c_2 = 0$$

$$c_2 = -2$$

$$c_1 = 1 - (-2) = 3$$

L sning

$$y(t) = 3e^{-2t} - 2e^{3t} + t e^{-2t}$$

Oppg ve 10

L  V v re et indreproduktrom og l 

$$v_1 \neq 0 \text{ og } v_2 \neq 0$$

Vis dersom v_1 og v_2 er ortogonale, s  er de line rt uavhengige

Ortogonale vektorer

$$\langle v_1, v_2 \rangle = 0$$

Line rt uavhengige

$$c_1 v_1 + c_2 v_2 = 0$$

har bare l sningen

$$c_1 = 0, c_2 = 0$$

$$c_1 v_1 + c_2 v_2 = 0$$

$$\langle c_1 v_1 + c_2 v_2, v_1 \rangle = \langle 0, v_1 \rangle$$

$$c_1 \langle v_1, v_1 \rangle + c_2 \langle v_2, v_1 \rangle = 0$$

siden de er ortogonale er det lik 0

$$c_1 \langle v_1, v_1 \rangle = 0$$

$$\text{Siden } v_1 \neq 0 \Rightarrow \langle v_1, v_1 \rangle \neq 0 \Rightarrow c_1 = 0$$

Sett inn i den opprinnelige ligningen

$$c_1 v_1 + c_2 v_2 = 0$$

$$c_2 v_2 = 0$$

$$c_2 = 0$$

Dermed har vi $c_1 = c_2 = 0$

v_1 og v_2 er lineært uavhengige